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Physics IV Notes

Oscillations, waves, optics, and quantum physics

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Welcome to the **Physics IV** course!

This material is designed as fundamental support for the subject, focusing on the study of oscillations, waves, an introduction to optics, and the foundational concepts that led to the birth of quantum physics.

Through this interactive book, we will explore everything from the simplest oscillatory systems to the wave behavior of matter.

Course Syllabus

The course is structured into the following thematic blocks:

1. **Equilibrium, damping, and oscillations**
2. **Harmonic oscillation**
3. **Damped and forced harmonic oscillation**
4. **Coupled oscillators**
5. **Wave motion: standing waves**
6. **Wave motion: traveling waves**
7. **Wave motion: waves in multiple dimensions**
8. **Classical description of light**
9. **The birth of quantum physics**

Bibliography and Resources

To delve deeper into the course content, we recommend the following main bibliography:

- P.A. Tipler and G. Mosca, “*Physics for scientists and engineers*”, 6th ed. (W.H. Freeman, 2007). Part II Oscillations and waves.
- R.D. Knight, “*Physics for scientists and engineers: A strategic approach with modern physics*”, 5th ed. (Pearson, 2022). Part IV Oscillations and waves.
- R. Fitzpatrick, “*Oscillations and waves: An introduction*”, 2nd ed. (CRC Press, 2018).
- H. Georgi, “*The physics of waves*” (Prentice Hall, 1993).
- G.C. King, “*Vibrations and waves*”, 6th ed. (Wiley, 2009).
- H.J. Pain, “*The physics of vibrations and waves*”, 6th ed. (Wiley, 2005).
- A.P. French, “*Vibrations and waves*”, (CRC Press, 1971).
- F.S. Crawford, “*Waves (Berkeley physics course vol. 3)*”, (McGraw-Hill, 1968).

Additional Interactive Resources:

- General physics animations: surendranath.org
- Acoustics and vibrations animations: [D. Russell \(PSU\)](http://D.Russell(PSU))
- Math and physics applets: falstad.com
- MIT Mathlets: mathlets.org

- Animations for physics and astronomy (PSU and mrg3): phys23p.sl.psu.edu / YouTube
- PhET Interactive Simulations: phet.colorado.edu

BIBLIOGRAPHY